IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 14 August 2025, accepted 23 September 2025, date of publication 1 October 2025, date of current version 9 October 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3616770

RESEARCH ARTICLE

Domain-Adaptive Full-Face Gaze Estimation via Novel-View-Synthesis and Feature Disentanglement

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This work was supported by the Grants-in-Aid for Scientific Research of Japan Society for the Promotion of Science (JSPS) KAKENHI under Grant JP21K11932.

ABSTRACT Along with the recent development of deep neural networks, appearance-based gaze estimation has succeeded considerably when training and testing within the same domain. Compared to the within-domain task, the variance of different domains makes the cross-domain performance drop severely, preventing gaze estimation deployment in real-world applications. Among all the factors, the ranges of head pose and gaze are believed to play significant roles in the final performance of gaze estimation, while collecting large ranges of data is expensive. This work proposes an effective model training pipeline consisting of a training data synthesis and a gaze estimation model for unsupervised domain adaptation. The proposed data synthesis leverages the single-image 3D reconstruction to expand the range of the head poses from the source domain without requiring a 3D facial shape dataset. To bridge the inevitable gap between synthetic and real images, we further propose an unsupervised domain adaptation method suitable for synthetic full-face data. We propose a disentangling autoencoder network to separate gaze-related features and introduce background augmentation consistency loss to utilize the characteristics of the synthetic source domain. Through comprehensive experiments, it shows that the model using only our synthetic training data can perform comparably to real data extended with a large label range. Our proposed domain adaptation approach further improves the performance on multiple target domains. The data are available at https://github.com/ut-vision/Gaze-NV-Rendering

INDEX TERMS Data synthesis, domain adaptation, gaze estimation.

I. INTRODUCTION

Appearance-based gaze estimation is a promising solution to indicate human user attention in various settings with a single webcam as the input device, such as human-robot interaction [1], [2], social interaction [3], [4], and entertainment [5], [6]. Machine learning-based methods have been evolving from convolutional neural networks [7], [8], [9] to vision transformers (ViT) [10], [11] for more robust performance under in-the-wild usage settings. There is also a trend to include the face image instead of just the eye region [8], [12]. Despite their effectiveness, it is well known that data-driven methods are prone to overfitting to data bias and thus lose their generalization ability across environments, limiting their applicability in real-world scenarios.

In the research community, this issue has often been analyzed quantitatively in cross-domain evaluations with training and testing on different datasets [13]. There is a

The associate editor coordinating the review of this manuscript and approving it for publication was Li Zhang.

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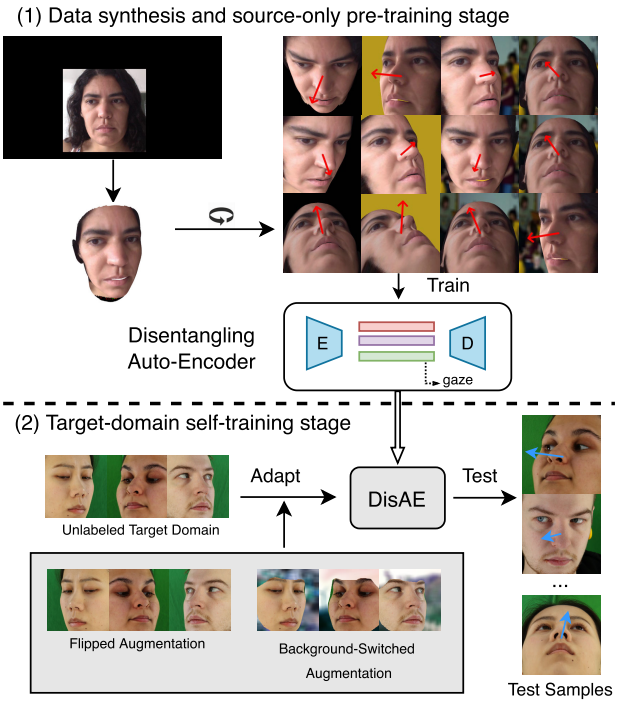


FIGURE 1. Overview of our approach with two stages. (1) With a monocular source image as input, we synthesize the data for a large range of head poses stemming from the 3D face reconstruction. We propose a feature disentangling auto-encoder network pre-trained only on the synthetic data from the source images. (2) For the unlabeled target domain, we leverage self-training to adapt the model to the unlabeled target domains.

significant performance drop when the trained model is tested on different domains with variations in personal appearances, head poses, gaze directions, and lighting conditions [9], [14], [15]. To improve the robustness of deep estimation models, many large-scale datasets have been collected and contributed to the community [9], [12], [13], [14], [16]. In-the-wild datasets provide diverse lighting [13], [14] and large subject scale [12], while controlled settings can capture images with extreme head poses and gaze directions [9]. However, acquiring gaze labels is costly, and constructing a comprehensive training dataset covering all conditions is not trivial.

Another line of work has created synthetic data as additional training data [17], [18], [19], [20]. Although synthetic images from gaze redirection can be used to augment existing training data [18], the label accuracy from learning-based generative models is not sufficient to serve as training data alone, especially for large angles. Learning-by-synthesis using 3D graphics has been proven effective in eye-only gaze estimation [17], [19], [21]. However, the domain gap between real and synthetic images is difficult to bridge [22], and no effective pipeline has been proposed to synthesize and adapt full-face images as training data.

In this work, we propose tackling the goal of generalizable appearance-based gaze estimation by leveraging data synthesis and the domain adaptation approach. As shown at the top of Fig. 1, our method forms a unified and practical pipeline of data synthesis and unsupervised domain adaptation. We perform single-image 3D face reconstruction to synthesize data with extended head poses and gaze direction ranges. With the synthetic data, we propose a novel unsupervised domain adaptation framework that combines disentangled representation learning with a self-training strategy. Our proposed disentangling auto-encoder (DisAE) is first trained on the synthetic source domain to learn gaze representations that are expected to generalize better to unseen domains. The model is then trained on unlabeled target domains in a self-training manner [23], [24], [25], [26], [27], [28]. Based on the characteristics of our synthetic data, we further introduce a background-switching data augmentation consistency loss for synthetic-real domain adaptation. Experiments with multiple target datasets show that the proposed pipeline significantly improves performance compared to the source dataset before reconstruction. Moreover, our single-image face reconstruction approach applies to most real-world settings, while multi-view face reconstruction may serve as an upper bound in performance. We also analyze in detail how closely single-view reconstruction can approach the training accuracy of multi-view reconstruction.

This manuscript is based on our previous publication [29], and parts of the text and figures are reused from it. The major changes are as follows. First, we fully updated the model training pipeline by introducing DisAE and a self-training strategy for unsupervised domain adaptation. Second, we added an experiment using multi-view reconstruction data from the ETH-XGaze dataset [9] to analyze the upper-bound performance of synthetic training. This also resulted in recalibrated camera parameters for the ETH-XGaze dataset, facilitating future multi-view gaze estimation research.

In summary, the contributions of this work are threefold.

(i) We are the first to propose an approach using single-view 3D face reconstruction to create synthetic training data for appearance-based gaze estimation. We further utilize the properties of the synthetic data to perform background switching for image appearance augmentation.

(ii) We propose a novel unsupervised domain adaptation approach that combines feature disentanglement with a self-training strategy. Experiments show that the proposed method is particularly effective in addressing synthetic-real domain gaps.

(iii) We provide a detailed comparison with multi-view face reconstruction to analyze the performance of the single-view approach. We also release the recalibrated camera extrinsic parameters for the ETH-XGaze dataset to facilitate further research.

II. RELATED WORKS

A. APPEARANCE-BASED GAZE ESTIMATION

While traditional model-based gaze estimation relies on 3D eyeball models and geometric features [30], [31], appearance-based methods [32] use a direct mapping from the image to

the gaze direction, enabling their use in a wider range of settings with less hardware dependency. Previous work on appearance-based gaze estimation has mostly used single-eye [17], [19], [32], [33], [34], [35], [36] or two-eye images as input [37], [38], [39], [40]. More recent works using full-face input [8], [12], [41], [42] have demonstrated higher robustness and accuracy in gaze estimation compared to eye-only methods.

To alleviate the data hunger of deep learning methods, multiple datasets have been proposed. Most gaze datasets are collected in indoor environments that lack diverse lighting conditions [16], [43], [44], [45]. Later works switched to *in-the-wild* data collection to capture varied lighting conditions [8], [13]. However, these datasets have limited ranges of head pose and gaze directions due to the data collection devices, such as laptops [8], [13], cellphones [12], and tablets [12], [45]. More recent datasets have further extended diversity in head pose and environmental conditions [9], [14]. Nevertheless, acquiring training datasets that meet the requirements for head pose and appearance variations in deployment environments still demands significant effort. In this work, we extend the head pose ranges of source datasets with full-face synthetic data for the gaze estimation task.

B. LEARNING-BY-SYNTHESIS FOR GAZE ESTIMATION

Previous studies have created synthetic training data for the gaze estimation task to bypass the burden of real-world data collection. One direction is to use multi-view stereo reconstruction [19]. However, the multi-view setup has the drawback that the environment is restricted to laboratory conditions. Another group of methods used hand-crafted computer graphics models to generate samples with arbitrary head poses, gaze directions, and lighting conditions [21], [46]. Unfortunately, these synthesized images from graphics models have a non-negligible domain gap compared to real images. Gaze redirection has also been proposed to generate synthetic data for personal gaze estimator training [17], [18], [47]. However, these approaches cannot guarantee that the generated samples exactly match the target gaze label. Alternatively, this work employs a single-image 3D face reconstruction approach for accurate data synthesis, enabling the generation of synthetic training data with higher realism and precision than previous methods.

C. DOMAIN GAP IN GAZE ESTIMATION

The cross-domain gap is a significant challenge in appearance-based gaze estimation. To address this, previous works have either improved generalizability by devising better gaze representation learning from the source domain [36], [39], [48] or directly leveraged target domain data with unsupervised domain adaptation [27], [40], [49]. For instance, the disentangling transforming encoder-decoder [39] separates features to obtain more domain-invariant gaze representations. PureGaze [48] extracts purified gaze features from image features using a self-adversarial learning strategy. Nevertheless, domain generalization in gaze estimation remains challenging due to numerous influencing factors. In this work, we focus on synthesizing data tailored to the head pose distribution of the target domain and primarily emphasize adaptation rather than generalization.

Unsupervised domain adaptation has succeeded in tasks such as classification and segmentation [50], [51], [52], but only limited work has focused on regression tasks [25], [26], [53], [54], among which gaze estimation is particularly challenging. SimGAN [22] adapts synthetic training data to resemble real target images before training, while more recent methods focus on directly adapting the model to the target domain using self-supervised learning [49], [55], [56], [57]. Liu et al. [27] proposed a framework using collaborative learning that adapts to the target domain with only very few images guided by outlier samples. Other methods have targeted specific issues, such as jitter samples [58] and in-plane rotation inconsistency [59], and developed specialized self-supervised learning strategies to address them. GazeCLR [60] leveraged multi-view consistency as the underlying principle of its contrastive learning framework. Wang et al. [61] proposed a contrastive learning framework based on the assumption of similarity between gaze labels and features. LatentGaze [49] employed generative adversarial networks to transform the target domain into the source domain for easier estimation. In summary, most previous work has focused on adapting a wide range to a narrow range, and the effectiveness of synthetic source data has not been evaluated. Thus, we propose a gaze estimation model that explicitly learns more robust gaze representations from synthetic data and adapts to the real domain using unlabeled data.

D. 3D FACE RECONSTRUCTION

There has been significant progress in monocular 3D face reconstruction techniques in recent years [62]. Although reconstructed 3D faces have also been used to augment face recognition training data [63], [64], [65], no prior work has yet explored their usage in full-face appearance-based gaze estimation. Most methods based on 3D morphable models [66], [67] approximate facial textures via appearance bases [68], [69], [70], where the appearance of the eye region can be distorted. To preserve accurate gaze labels after reconstruction, our proposed data synthesis approach employs 3D face reconstruction methods that sample textures directly from the input image [71], [72], [73], [74], [75], [76], [77]. In addition, since many prior works rely on orthogonal or weak perspective projection models, we also investigate how to precisely align the reconstruction results with the source camera coordinate system.

Beyond monocular-based methods, there are multi-view stereo approaches that produce higher-quality 3D geometry [78], as well as Neural Radiance Field (NeRF) methods that represent the scene implicitly as a radiance field to

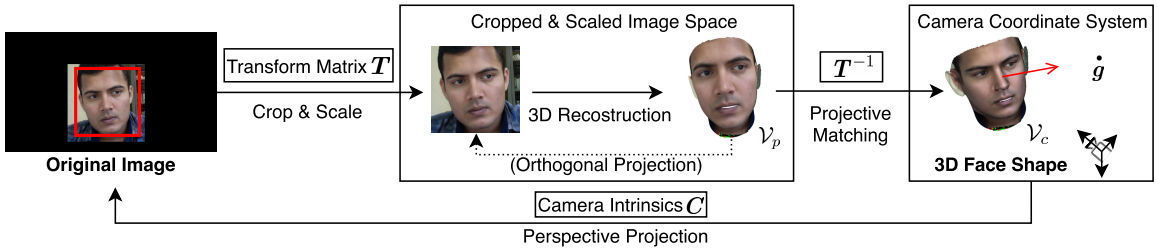


FIGURE 2. Overview of the data synthesis pipeline. Given monocular images as input, we first obtain a face patch by cropping and scaling. A 3D face model is then fitted to the input patch. Since 3D face reconstruction methods typically generate facial meshes under an orthogonal projection model, we introduce a projective matching step to convert the mesh from the image-pixel system to the physical camera coordinate system. After this process, the 3D face is aligned with the ground-truth gaze position, enabling rotation of the 3D face to simulate different head poses.

achieve fine-level details. However, these methods require long processing times and a large number of views, and they are also sensitive to low-quality images [79].

III. NOVEL-VIEW SYNTHESIS

Our proposed method consists of two parts: data synthesis and synthetic-real domain adaptation. In this section, we introduce the data synthesis process with 3D face reconstruction to generate samples with a large range of head poses.

A. OVERVIEW

Fig. 2 shows the overview of our data synthesis pipeline. Given an ordinary single-view gaze dataset, we apply 3D face reconstruction on each sample to synthesize face images with novel head poses while preserving accurate gaze direction annotations. We adopt a simple 3D face reconstruction to create the 3D face mesh in the pixel coordinate system, and propose a transformation process, named *projective matching*, to obtain the 3D face mesh in the camera coordinate system. Finally, 2D face images can be rendered using camera perspective projection with the 3D face mesh.

B. 3D FACE RECONSTRUCTION

The synthesis image generation requires the source gaze dataset, which consists of 1) face images, 2) the projection matrix (intrinsic parameters) $\boldsymbol{C}$ of the camera, and 3) the 3D gaze target position $\boldsymbol{g} \in \mathbb{R}^3$ in the camera coordinate system. Most of the existing gaze datasets satisfy our requirements [9], [12], [34], and yaw-pitch annotations can also be converted assuming a distance to the dummy target [14]. State-of-the-art learning-based 3D face reconstruction methods usually take a cropped face patch as input and output a 3D facial mesh, which is associated with the input image in an orthographic projection way. Without loss of generality, we assume that the face reconstruction method takes a face bounding box defined with center (c_x, c_y), width w_b, and height h_b in pixels and then resized to a fixed input size by factor (s_x, s_y). The reconstructed facial mesh is defined as a group of N vertices $\mathcal{V}_p = \{\boldsymbol{v}_p^{(i)}\}_{i=0}^N$. Each vertex is represented as $\boldsymbol{v}_p^{(i)} = [u^{(i)}, v^{(i)}, d^{(i)}]^\top$ in the right-handed coordinate system, where u and v directly correspond to the pixel locations in the input face patch and d is the distance to the u-v plane in the same pixel unit. This representation has been used by recent works [72], [73], [76], [80], [81], and we can convert an arbitrary 3D representation to it by projecting the reconstructed 3D face onto the input face patch.

Our goal is to convert the vertices of the reconstructed 3D face $\mathcal{V}_p$ to another 3D representation $\mathcal{V}_c = \{\boldsymbol{v}_c^{(i)}\}_{i=0}^N$ where each vertex $\boldsymbol{v}_c^{(i)} = [x^{(i)}, y^{(i)}, z^{(i)}]^\top$ is in the original camera coordinate system so that it can be associated with the gaze annotation $\boldsymbol{g}$. In this way, the gaze target location can also be represented in the facial mesh coordinate system, and we can render the facial mesh under arbitrary head or camera poses together with the ground-truth gaze direction information.

C. PROJECTIVE MATCHING

Projective matching, in a nutshell, is to approximate parameters for transforming the $\mathcal{V}_p$ to $\mathcal{V}_c$ such that $\mathcal{V}_c$ matches the perspective projection.

In detail, since u and v of each reconstructed vertex $\boldsymbol{v}_p$ are assumed to be aligned with the face patch coordinate system, $\boldsymbol{v}_c$ must be on the back-projected ray as

$$\boldsymbol{v}_c = \lambda \frac{\boldsymbol{C}^{-1}\boldsymbol{p}_o}{||\boldsymbol{C}^{-1}\boldsymbol{p}_o||} = \lambda \frac{\boldsymbol{C}^{-1}\boldsymbol{T}^{-1}\boldsymbol{p}}{||\boldsymbol{C}^{-1}\boldsymbol{T}^{-1}\boldsymbol{p}||}, \tag{1}$$

where $\boldsymbol{p}_o = [u_o, v_o, 1]^\top$ and $\boldsymbol{p} = [u, v, 1]^\top$ indicates the pixel locations in the original image and the face patch in the homogeneous coordinate system, respectively, and

$$\boldsymbol{T} = \begin{bmatrix} s_x & 0 & -s_x(c_x - \frac{w}{2}) \\ 0 & s_y & -s_y(c_y - \frac{h}{2}) \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

represents the cropping and resizing operation to create the face patch, *i.e.*, $\boldsymbol{p} = \boldsymbol{T}\boldsymbol{p}_o$. The scalar λ indicates scaling along the back-projection ray and physically means the distance between the camera origin and $\boldsymbol{v}_c$.

Since Eq. (1) does not explain anything about d, our task can be understood as finding λ, which also maintains the relationship between u, v, and d. Therefore, as illustrated in Fig. 3, we propose to define λ as a function of d as $\lambda = \alpha d + \beta$. α indicates a scaling factor from the pixel to physical (*e.g.*, millimeter) unit, and β is the bias term to align

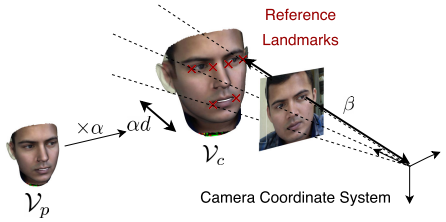


FIGURE 3. Determining the location of $\mathcal{V}_c$ via parameters α and β. α indicates a scaling factor from the pixel to physical (*e.g.*, millimeter) unit, and β is the bias term to align αd to the camera coordinate system.

αd with the camera coordinate system. Please note that α and β are constant parameters determined for each input image and applied to all vertices from the same image.

We first fix α based on the distance between two eye centers (midpoints of two eye corner landmarks) compared to a physical reference 3D face model. 3D face reconstruction methods usually require facial landmark detection as a pre-processing step. Thus, we can naturally assume that the corresponding vertices in $\mathcal{V}_p$ to the eye corner landmarks are known. We use a 3D face model with 68 landmarks (taken from the OpenFace library [82]) as our reference. We set $\alpha = l_r/l_p$, where l_p and l_r are the eye-center distances in $\mathcal{V}_p$ and in the reference model, respectively.

We then determine β by aligning the reference landmark depth in the camera coordinate system. In this work, we use the face center as a reference, which is defined as the centroid of the eyes and the mouth corner landmarks, following previous works on full-face gaze estimation [8], [83]. We use the same face center as the origin of the gaze vector through the data normalization and the gaze estimation task.

We approximate β as the distance between the ground-truth 3D reference location and the scaled/reconstructed location as $\beta = ||\bar{\boldsymbol{v}}|| - \alpha\bar{d}$. $\bar{d}$ is the reconstructed depth values computed as the mean of six landmark vertices corresponding to the eye and mouth corner obtained in a similar way as when computing α. $\bar{\boldsymbol{v}}$ is the centroid of the 3D locations of the same six landmarks in the camera coordinate system, which are obtained by minimizing the projection error of the reference 3D model to the 2D landmark locations using the Perspective-n-Point (PnP) algorithm [84].

D. TRAINING DATA SYNTHESIS

With the 3D face mesh $\mathcal{V}_c$ in the original camera coordinate system, we can render 2D face images under arbitrary head poses with the ground-truth gaze vector. To render a face image in a new camera coordinate system defined by the extrinsic parameters $\boldsymbol{R}_e, \boldsymbol{t}_e$, we project the vertex $\boldsymbol{v}_c$ and gaze target position $\boldsymbol{g}$ onto the new system by applying the transformation $\boldsymbol{R}_e\boldsymbol{v}_c + \boldsymbol{t}_e$ and $\boldsymbol{R}_e\boldsymbol{g} + \boldsymbol{t}_e$, respectively. To render a face image from a source head pose $\boldsymbol{R}_s, \boldsymbol{t}_s$ to a target head pose $\boldsymbol{R}_t, \boldsymbol{t}_t$, we transform the vertices and gaze position by applying the transformation $\boldsymbol{R}_t(\boldsymbol{R}_s)^{-1}(\boldsymbol{v}_c - \boldsymbol{t}_s) + \boldsymbol{t}_t$, similar for $\boldsymbol{g}$.

Except for the geometric augmentation, we further augment the data with image appearances in terms of lighting conditions and background appearances by virtue of the flexible synthetic rendering. Although most 3D face reconstruction methods do not reconstruct lighting and albedo, we maximize the diversity of rendered images by controlling the global illumination. In the PyTorch3D renderer, the ambient color $[r, g, b]$ represents the ambient light intensity, ranging from 0 to 1, in which 1 is the default value for full lighting. For weak-light images, we set them to be a random value between 0.25 and 0.75. We set the background to random colors or scenes by modifying the blending settings. Random scene images are taken from the Places365 dataset [85], and we apply blurring to them before rendering faces. Overall, among all generated images, the ratio of black, random color, and random scene is set to 1:1:3, and half of them are weak lighting. Fig. 4 shows examples of the synthesized images using MPIIFaceGaze [13] and ETH-XGaze [9].

In the experiments, we applied 3DDFA [76] to reconstruct 3D faces from the source dataset. After projective matching, we rendered new images using the PyTorch3D library [86].

IV. DOMAIN ADAPTATION WITH FEATURE DISENTANGLEMENT

Our synthetic dataset generation pipeline can render realistic face regions with accurate gaze labels, and these generated samples could be directly used to train a model for the cross-domain task. However, there remains an image appearance domain gap between the synthetic and real samples. In particular, the influence of the background, hair, clothing, and other non-face areas of the synthesized images on the gaze estimation model cannot be ignored. To address this gap, we propose a gaze estimation framework that adapts to the target domain through unsupervised domain adaptation. However, gaze-unrelated features, such as image appearance and head pose, can make the adaptation unstable [29], [49]. To mitigate this issue, we first devise the disentangling auto-encoder (DisAE) (Sec. IV-A) to separate gaze-related features during supervised training on the source domain. We then adapt DisAE to the target domain using a self-training approach [25], [26]. Since our synthetic source domain includes random backgrounds (Sec. III-D), we propose a background-switching consistency loss on the target domain as one of the self-training objectives.

A. DISENTANGLING AUTO-ENCODER

As the base architecture for adaptation, we propose a disentangling auto-encoder to separate gaze-unrelated features and reduce their influence on subsequent self-training. Specifically, as shown at the top of Fig. 5, we propose an encoder-decoder architecture to disentangle appearance, head, and gaze embeddings. For prediction, an MLP is used to estimate head pose $\hat{\boldsymbol{h}}$, and a vision transformer is used to predict gaze direction $\hat{\boldsymbol{g}}$. Finally, all features are concatenated and fed into a decoder to reconstruct the

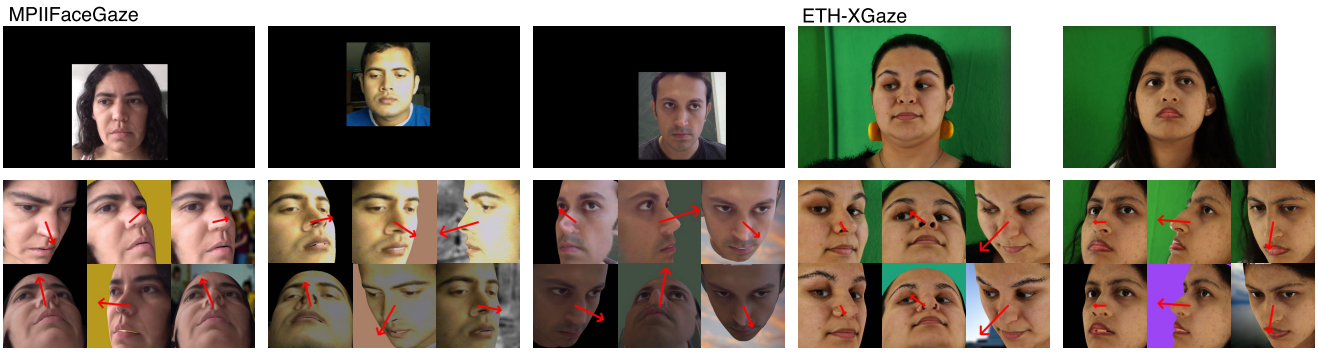


FIGURE 4. Examples of the synthesized images. The first row shows the source images from MPIIFaceGaze [8] and ETH-XGaze [9] datasets. For MPIIFaceGaze, the second and third rows show synthesized images in full and weak lighting. For ETH-XGaze, the second row shows the real images from the dataset, and the third row shows our synthetic images with the same head poses as the real samples. For each synthetic example, the three columns show the black, color, and scene background in turn. The red arrows indicate gaze direction vectors.

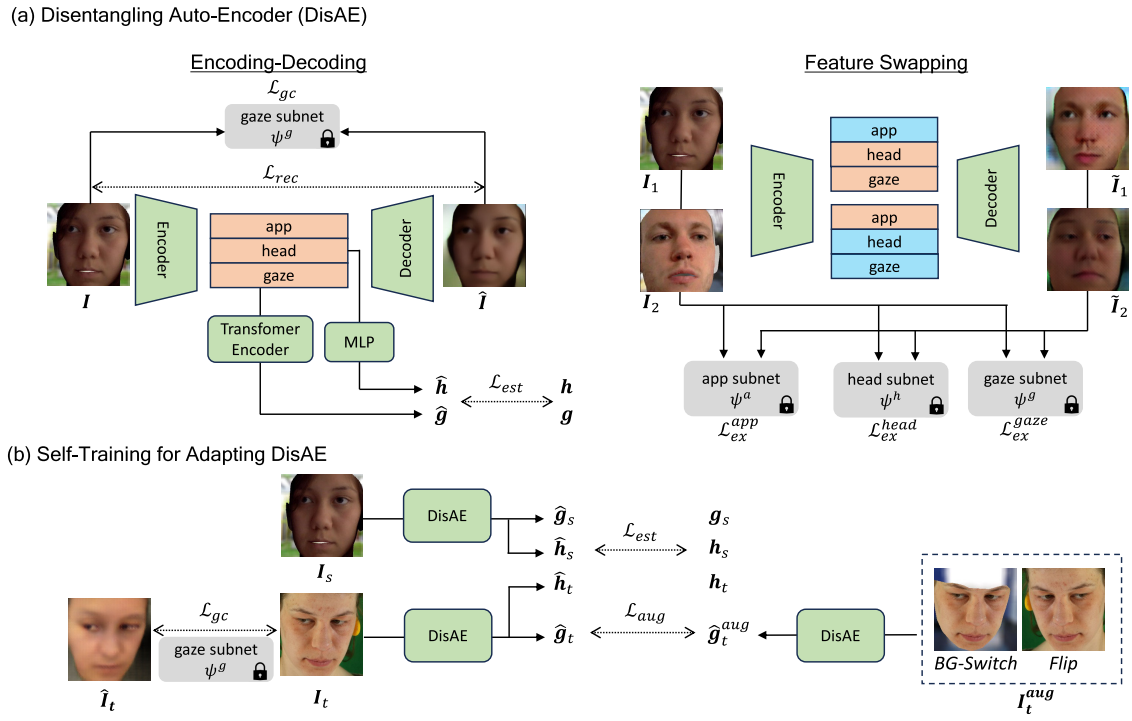


FIGURE 5. Overview of our synthetic-real domain adaptation approach. Top: The encoder-decoder structure of DisAE for feature disentanglement. Three subnets $\psi^{\mathbf{a}}$, $\psi^{\mathbf{h}}$, and $\psi^{\mathbf{g}}$ are used to disentangle appearance, head, and gaze features, respectively. Gaze features are fed into a vision transformer to predict gaze direction $\hat{g}$, while head features are fed into an MLP to predict head pose $\hat{h}$. Bottom: Augmentation consistency is applied during unsupervised domain adaptation of DisAE to the target domain.

image. After disentanglement, the extracted features become more correlated with gaze while reducing the impact of unrelated factors, making the pre-trained model better suited for domain adaptation.

To achieve this disentanglement, we prepare three additional subnets: a face recognition network ψ^{a} that predicts appearance embeddings, ψ^{h} for predicting head pose, and ψ^{g} for predicting gaze, all based on a ResNet-18 architecture. These three subnets are first trained on the source domain, after which we train DisAE using the losses derived from them. Note that only synthetic source-domain data are used to pre-train DisAE. As shown at the top of Fig. 5, the pre-training loss mainly consists of two components: *encoding-decoding* and *feature swapping*.

1) ENCODING-DECODING

Encoding-decoding losses are defined between each input image and the output (decoded image and estimation result) from the DisAE architecture. **Reconstruction loss** is defined as $\mathcal{L}_{\mathrm{rec}} = |\boldsymbol{I} - \hat{\boldsymbol{I}}|_1$, where $\hat{\boldsymbol{I}}$ is the reconstructed image and $\boldsymbol{I}$ is the ground-truth image. **Estimator loss** is the commonly

used ℓ_1 loss on the gaze labels $\boldsymbol{g}$, defined in terms of the pitch and yaw dimensions. In addition to the gaze loss, we also calculate the head pose loss using the dataset's head pose label $\boldsymbol{h}$. Taken together, the estimator loss is defined as

$$\mathcal{L}_{\text{est}} = |\boldsymbol{g} - \hat{\boldsymbol{g}}|_1 + \lambda_{\text{head}}|\boldsymbol{h} - \hat{\boldsymbol{h}}|_1,$$

where $\hat{\boldsymbol{g}}$ and $\hat{\boldsymbol{h}}$ are the predicted gaze and head directions.

Gaze consistency loss is designed to make the gaze features insensitive to variations in appearance features. We add random noise $\mathcal{N}(\mathbf{0}, \mathbf{0.1})$ to the appearance features before feeding them into the decoder. The reconstructed image $\hat{\boldsymbol{I}}$ with noise is expected to have the same gaze direction as the original image $\boldsymbol{I}$. Therefore, we use the pre-trained gaze subnet ψ^{g} to compute the gaze consistency loss between the two images as

$$\mathcal{L}_{\text{gc}} = \left|\psi^{\text{g}}(\boldsymbol{I}) - \psi^{\text{g}}(\hat{\boldsymbol{I}})\right|_1.$$

2) FEATURE SWAPPING

Feature swapping losses are defined between image pairs whose disentangled features are swapped after passing through the encoder. **Feature exchange consistency loss** is introduced to encourage the disentangling of face appearance from gaze and head pose features. Specifically, we swap the appearance embeddings between two samples $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$ and decode them into $\tilde{\boldsymbol{I}}_1$ and $\tilde{\boldsymbol{I}}_2$. We ensure that $\tilde{\boldsymbol{I}}_1$ retains the head pose and gaze features of $\boldsymbol{I}_1$ while adopting the appearance features of $\boldsymbol{I}_2$. Similarly, $\tilde{\boldsymbol{I}}_2$ retains the head pose and gaze features of $\boldsymbol{I}_2$ while adopting the appearance features of $\boldsymbol{I}_1$. This process is illustrated at the top of Fig. 5, and the loss is formulated as

$$\mathcal{L}_{\text{ex}}(\boldsymbol{I}_1, \boldsymbol{I}_2) = \sum_{\psi} \left|\psi(\boldsymbol{I}_1) - \psi(\tilde{\boldsymbol{I}}_1)\right|_1 + \left|\psi(\boldsymbol{I}_2) - \psi(\tilde{\boldsymbol{I}}_2)\right|_1. \tag{3}$$

The total loss for pre-training DisAE is the sum of all the above losses:

$$\mathcal{L}_{\text{source}} = \mathcal{L}_{\text{est}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{gc}}\mathcal{L}_{\text{gc}} + \lambda_{\text{ex}}\mathcal{L}_{\text{ex}}. \tag{4}$$

B. SELF-TRAINING ON TARGET DOMAINS

After the source-domain-only supervised training of DisAE, we leverage unlabeled target-domain data using an augmentation consistency loss, inspired by the widespread use of data augmentation in self-training [25], [26], [27]. Briefly, we apply data augmentations to unlabeled target-domain images and enforce the model to output the same gaze direction for both the original and augmented samples. These augmentations tune the gaze-related features in DisAE for the target domain without requiring gaze labels.

The augmentation consistency loss is defined as the ℓ_1 loss between the gaze prediction of the original target image and that of the augmented images:

$$\mathcal{L}_{\text{aug}} = \lambda_{\text{bg}}\mathcal{L}_{\text{bg}} + \lambda_{\text{flip}}\mathcal{L}_{\text{flip}},$$

where $\mathcal{L}_{\text{bg}} = \left|\hat{\boldsymbol{g}}_t - \hat{\boldsymbol{g}}_t^{\text{bg}}\right|_1$ and $\mathcal{L}_{\text{flip}} = \left|\hat{\boldsymbol{g}}_t - \hat{\boldsymbol{g}}_t^{\text{flip}}\right|_1$.

As one of the augmentations, we introduce a background-switching loss, particularly tailored for our synthetic data. Specifically, we obtain a facial region mask from detected landmarks and replace the background with random images from the Places365 dataset [85], which should not affect the gaze direction. Since our synthetic source domain already contains random background regions, there is no gaze-relevant information in non-face areas, and the pre-trained DisAE is expected to focus on facial regions. By training the model on the target domain so that gaze estimation remains consistent across images with swapped backgrounds, the model can leverage this property for adaptation. Because most geometric augmentations may alter gaze estimation, we only consider horizontal flipping as another augmentation. In this case, the target image is flipped horizontally, and the yaw value of the gaze label is negated.

In addition to the augmentation consistency loss, we employ similar losses as in the pre-training process. We continue to compute the same gaze consistency loss $\mathcal{L}_{\text{gc}} = \left|\psi^{\text{g}}(\boldsymbol{I}_t) - \psi^{\text{g}}(\hat{\boldsymbol{I}}_t)\right|_1$ for the target domain. Since head pose labels can be obtained through the data normalization process, even for unlabeled target-domain images, we define the estimator loss as $\mathcal{L}_{\text{est}} = \left|\boldsymbol{g}_s - \hat{\boldsymbol{g}}_s\right|_1 + \lambda_{\text{head}}\left(\left|\boldsymbol{h}_s - \hat{\boldsymbol{h}}_s\right|_1 + \left|\boldsymbol{h}_t - \hat{\boldsymbol{h}}_t\right|_1\right)$, where $\hat{\boldsymbol{g}}_s$ and $\hat{\boldsymbol{h}}_s$ are the predicted gaze and head directions from the source-domain input $\boldsymbol{I}_s$, and $\hat{\boldsymbol{h}}_t$ is the predicted head direction from the target-domain input $\boldsymbol{I}_t$.

Consequently, the total loss for the adaptation stage is defined as

$$\mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{est}} + \lambda_{\text{gc}}\mathcal{L}_{\text{gc}} + \mathcal{L}_{\text{aug}}. \tag{5}$$

C. IMPLEMENTATION DETAILS

The face recognition subnet ψ^a is trained using a triplet loss [87], and the estimation subnets ψ^h and ψ^g are both trained using ℓ_1 loss. In the source-only training stage of DisAE, all subnets ψ^a, ψ^h, and ψ^g are trained in a fully supervised manner with source-domain data. In the target-domain adaptation stage, target-domain data are also used for fully supervised training of the face recognition subnet ψ^a and head subnet ψ^h, while the gaze subnet ψ^g remains the same as trained on the source domain. We train DisAE using the Adam optimizer [88] for 12 epochs, with a learning rate of 0.001 decayed by a factor of 0.1 every five epochs. In addition to the target sample flipping described in Sec. IV-B, we also horizontally flip all images in the source training set to alleviate inconsistent accuracy between horizontally symmetric images.

During adaptation, we randomly sample 2,000 examples from the target-domain datasets and adapt the model for ten epochs. The final result is reported as the average over five random-seed repetitions. For the coefficients, we empirically set $\lambda_{\text{head}} = 0.5$, $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{ex}} = 1.0$, and $\lambda_{\text{gc}} = 1.0$. For the augmentation consistency losses, the DisAE at the bottom of Fig. 5 shares the weights across augmentations.

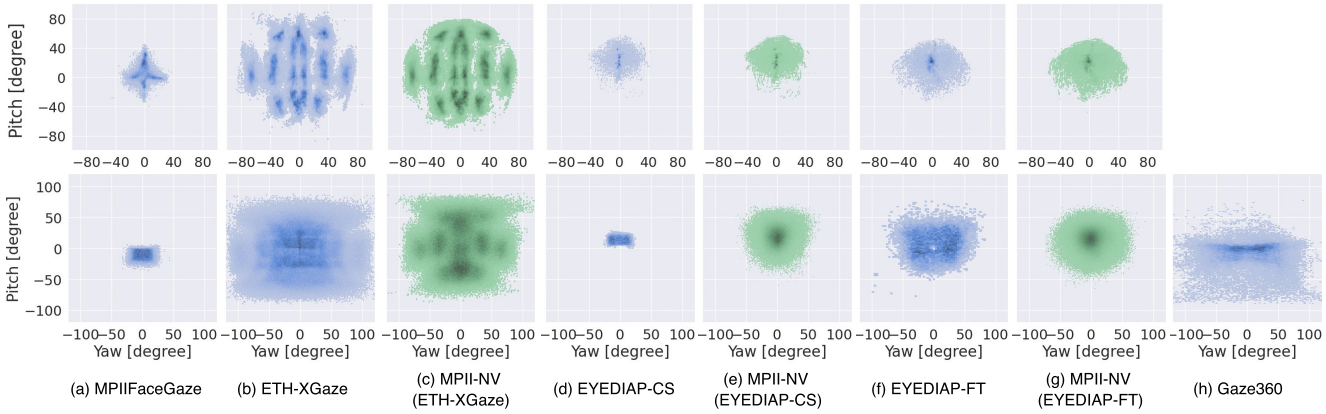


FIGURE 6. Head pose (top row) and gaze direction (bottom row) distributions of original datasets and our synthetic data. We always take the MPIIFaceGaze (a) as the source dataset to extend for the distribution of another dataset. By taking ETH-XGaze (b) as the target, we synthesize the data MPII-NV (ETH-XGaze) (c). By taking EYEDIAP (CS) (d) as the target, we synthesize MPII-NV (EYEDIAP-CS) (e). By taking EYEDIAP (FT) (f) as the target, we synthesize the MPII-NV (EYEDIAP-FT) (g). The last column shows the gaze distribution of Gaze360 (h) as a reference, which does not provide the head pose distribution.

Each augmentation loss is computed separately and summed, with $\lambda_{\text{bg}} = 0.5$ and $\lambda_{\text{flip}} = 0.5$.

V. EXPERIMENTS

We conduct data extrapolation experiments (Sec. V-A) to demonstrate the angle extension enabled by our proposed data synthesis, as well as the adaptation performance of DisAE with ablation studies. Additionally, we compare multiple 3D face reconstruction methods in terms of their impact on the final gaze estimation performance when used as training data (Sec. V-C).

EXPERIMENTAL SETTINGS

1) DATASETS

MPIIFaceGaze [8] consists of over 38,000 images of 15 subjects with variant lighting conditions. Since we only use this dataset for data synthesis, we selected images with frontal head poses where both pitch and yaw angles of the head pose are smaller than 15°. To ensure that the number of samples from each subject is balanced for training, we randomly down-sampled or up-sampled the number of images of each subject to 1,500. **ETH-XGaze** [9] contains over one million images of 110 subjects under variant head poses. We follow the official evaluation protocol and use the public evaluation server to retrieve the test results. **EYEDIAP** [16] consists of over four hours of video data, using continuous screen targets (CS) or 3D floating object targets (FT). We treated the screen target and floating target subsets separately and sampled one image every five frames from the VGA videos using the pre-processing provided by Park et al. [39]. **Gaze360** [14] consists of indoor and outdoor images of 238 subjects with wide ranges of head poses and gaze directions. We followed the pre-processing of Cheng et al. [89], which excluded cases with invisible eyes, resulting in 84,902 images.

We apply the data normalization scheme commonly used in appearance-based gaze estimation [9], [83] for all datasets. We also directly render the 3D facial mesh in the normalized camera space. Unless otherwise noted, we follow the ETH-XGaze dataset [9] by setting the virtual camera focal length to 960 mm and the distance from the camera origin to the face center to 300 mm. Face images are rendered at 448×448 pixels and downscaled to 224×224 pixels before being fed into CNNs. 3D head pose is obtained by fitting a 6-landmark 3D face model to the 2D landmark locations provided by the datasets using the PnP algorithm [84].

2) BASELINE METHODS

We compare our method with several state-of-the-art gaze estimation approaches. In addition to the simple yet strong baseline **ResNet** [90], we include **Gaze-TR** [11], one of the state-of-the-art backbone architectures for single-image gaze estimation. It first extracts gaze features using ResNet, then feeds the feature maps into a transformer encoder, followed by an MLP to output gaze directions. **PureGaze** [48] extracts image features with ResNet, followed by an MLP for gaze estimation and decoding blocks for image reconstruction. **PnP-GA** [27] is a domain adaptation model based on the Mean-Teacher structure [91] and can be applied to many existing architectures. We follow the original implementation, which uses only 10 target samples for adaptation. **DANN** [92] incorporates a gradient reversal layer and a domain classifier on top of the backbone, forcing the model to learn invariant features from both source and target domains.

A. DATA EXTRAPOLATION

We explore the most practical setting for data extrapolation, *i.e.*, extending head poses and gaze directions from smaller ranges using synthesized data samples. We extend the source MPIIFaceGaze dataset to match the head pose distributions of

TABLE 1. Comparison of gaze estimation errors in degrees. From left to right, the columns list the training sets, model architectures, and test sets. All models trained on our synthesized dataset MPII-NV achieve the best performances on both ETH-XGaze and EYEDIAP-CS compared to other training datasets.

Training Datasets	Model	ETH-XGaze		EYEDIAP	
		Train	Test	CS	FT
MPIIFaceGaze	ResNet18 [90]	31.96	32.62	13.02	23.01
ETH-XGaze Train		-	-	9.81	**13.81**
Gaze360		17.17	17.55	9.67	16.04
MPII-NV		**12.84**	**13.99**	**5.37**	17.04
MPIIFaceGaze	PureGaze18 [48]	32.30	32.82	12.09	22.65
ETH-XGaze Train		-	-	8.79	**12.86**
Gaze360		16.72	17.06	7.61	13.59
MPII-NV		**12.93**	**14.07**	**5.45**	16.22
MPIIFaceGaze	GazeTR18 [11]	29.62	30.16	14.21	24.41
ETH-XGaze Train		-	-	8.91	**12.61**
Gaze360		16.41	16.91	8.23	13.08
MPII-NV		**11.36**	**12.01**	**5.58**	14.70

TABLE 2. Comparison of our method with baseline methods. The top block is source-domain-only methods, and the bottom block is methods that utilize unlabeled target-domain data.

Model	ETH-XGaze		EYEDIAP	
	Train	Test	CS	FT
ResNet18	12.84	13.99	5.37	17.04
PureGaze18	12.93	14.07	5.45	16.22
Gaze-TR18	11.36	12.01	5.58	14.70
DisAE (ours)	**11.21**	**12.00**	**5.22**	**13.50**
Res18 + PnP-GA	12.43	15.33	4.78	17.00
Res18 + DANN	15.32	15.51	6.93	15.00
Res18 + aug	15.13	16.82	5.45	16.29
DisAE + DANN	11.13	11.92	5.20	13.50
DisAE + aug (ours)	**10.99**	**11.89**	**4.63**	**12.69**

the target ETH-XGaze and EYEDIAP datasets, respectively. Note that we use the training set of ETH-XGaze as the reference for the target head pose distribution. Head pose values are obtained through the data normalization process, and each source image is reconstructed and rendered with 16 new head poses randomly selected from the target dataset. To avoid extreme profile views with fully occluded eyes, we discard cases where the pitch-yaw vector's ℓ_2-norm exceeds 80° during data synthesis. As a result, MPIIFaceGaze is extended to three synthetic datasets—ETH-XGaze, EYEDIAP CS, and EYEDIAP FT—each containing 360,000 images. We refer to these datasets collectively as MPII-NV.

1) COMPARISON OF DATASETS

We first evaluate how our data synthesis approach improves performance compared to other baseline training datasets. As a real-image baseline, we use the Gaze360 [14] dataset, which covers a wide gaze range. The head pose and gaze distributions of the source and target real datasets (blue) and the synthetic datasets (green) are shown in Fig. 6, along with the gaze distribution of Gaze360 (head pose is not provided). Since we synthesize data based on head pose distribution, the resulting gaze distribution does not exactly match the target but only roughly overlaps.

The comparison of training datasets under several baseline models is presented in Tab. 1, where numbers represent the angular error. We compare different state-of-the-art models in terms of gaze estimation performance when trained on different datasets. From the table, we observe that all models trained on our synthesized dataset MPII-NV achieve the best performance on both ETH-XGaze and EYEDIAP-CS compared to other training datasets. Note that MPII-NV is purely an extension of the original MPIIFaceGaze with larger head pose coverage. The significant improvements obtained by models trained on MPII-NV over MPIIFaceGaze indicate that our synthetic data pipeline can generate useful data for cross-dataset training. For EYEDIAP-FT, better performance was obtained when using real data from ETH-XGaze. One possible explanation is that EYEDIAP-FT exhibits a larger offset between gaze and head pose due to the use of physical gaze targets, such that our data, synthesized based on head pose, cannot fully reproduce the target gaze distribution (Fig. 6).

2) COMPARISON OF MODELS

Using the synthetic data, we evaluate the proposed DisAE under both cross-dataset and unsupervised domain adaptation settings. In Tab. 2, we fix MPII-NV as the training dataset to compare DisAE with state-of-the-art methods. The top block of Tab. 2 reports cross-dataset evaluation results without adaptation, *i.e.*, all methods are trained only on source-domain samples. We observe that DisAE outperforms the three baseline models across all test datasets, demonstrating the advantage of feature disentanglement even without domain adaptation. The bottom block of Tab. 2 shows results with domain adaptation using unlabeled samples from the target test sets. Here, our proposed DisAE successfully adapts

to all target domains, achieving the lowest estimation errors in the last row.

To evaluate the effectiveness of combining DisAE with our self-training strategy, we apply the same augmentation consistency adaptation to ResNet-18 and refer to it as *Res18 + aug* in Tab. 2. This baseline performs worse than DisAE, indicating that unsupervised domain adaptation is difficult to achieve with simple data augmentation due to gaze-unrelated features present in face images. In contrast, our proposed DisAE effectively alleviates this issue by focusing on gaze-related features, thereby enhancing the model's adaptability to the target domain. We further apply DANN on DisAE by feeding the disentangled gaze features into a gradient reversal layer and domain classifier to compute a domain classification loss. In *DisAE + DANN*, although DANN does not provide notable improvements on most test sets, DisAE demonstrates more stable adaptation performance compared to the basic ResNet.

B. ABLATION STUDIES

We first conduct an ablation study on source-domain data augmentation using image flipping. As shown in the top block of Tab. 3, DisAE achieves lower error than the non-flipping setting on all test datasets, demonstrating that flipping (effectively doubling) the training data is a simple yet valuable approach to address inconsistencies in symmetric images.

For the domain adaptation stage, we examine individual loss terms based on Eq. 5. Specifically, we separately evaluate the flipping augmentation loss ($\mathcal{L}_{\text{flip}}$) and the background replacement augmentation loss ($\mathcal{L}_{\text{bg}}$). From the table, we observe that all proposed losses gradually improve accuracy across most test datasets.

One exception is that the loss $\mathcal{L}_{\text{gc}}$ does not effectively contribute to EYEDIAP, possibly due to its low image resolution. It may also slightly interfere with the benefits of other loss terms, particularly on EYEDIAP FT. As mentioned previously, EYEDIAP FT exhibits a larger offset between gaze and head pose due to the use of physical gaze targets. This makes it more challenging for the model to generate reconstructed images with a consistent gaze direction. As a result, overemphasizing $\mathcal{L}_{\text{gc}}$ may harm overall gaze accuracy.

C. ANALYSIS OF DATA QUALITY

In this section, we compare face reconstruction performance between our proposed single-view approach and a more complex multi-view method. The primary goal of this comparison is to establish an upper bound on performance when using synthetic data for training gaze estimation models, given that multi-view synthetic data offers higher photorealism. In addition, we compare multiple single-view reconstruction methods to analyze the impact of reconstruction quality on model accuracy. Specifically, we reconstruct the frontal-camera images of ETH-XGaze [9] and rotate them to match the viewpoints of the other cameras, resulting in a synthetic version of ETH-XGaze, denoted as *XGazeF-NV*.

TABLE 3. Adaptation effect with respect to the individual loss.

	ETH-XGaze		EYEDIAP	
	Train	Test	CS	FT
DisAE w.o. flip	11.32	12.03	5.90	13.85
DisAE	11.21	12.00	5.22	13.50
$\mathcal{L}_{\text{gc}}$	11.10	11.90	5.22	13.43
$\mathcal{L}_{\text{bg}}$	11.12	12.00	4.96	11.54
$\mathcal{L}_{\text{flip}}$	11.40	12.21	5.36	11.40
$\mathcal{L}_{\text{bg}}+\mathcal{L}_{\text{flip}}$	11.32	12.12	4.81	**11.34**
$\mathcal{L}_{\text{gc}}+\mathcal{L}_{\text{bg}}$	11.03	11.99	4.97	13.00
$\mathcal{L}_{\text{gc}}+\mathcal{L}_{\text{flip}}$	11.09	11.95	4.80	12.94
$\mathcal{L}_{\text{gc}}+\mathcal{L}_{\text{bg}}+\mathcal{L}_{\text{flip}}$ (ours)	**10.99**	**11.89**	**4.63**	12.69

1) MULTI-VIEW RECONSTRUCTION

Since ETH-XGaze [9] is a camera-synchronized dataset, we implement multi-view reconstruction using Agisoft Metashape software [93]. Because reconstruction quality degrades significantly in dark environments, we only reconstruct the full-light frames.

Through preliminary analysis, we confirmed that there is a discrepancy between the external parameters provided by the dataset and the actual camera images. This is possibly due to drift in the camera positions after calibration, and the discrepancy varies across subjects. Therefore, we first use Metashape to optimize the camera extrinsic parameters for each subject. The Metashape optimization takes as input the 18 images of each frame of the target subject, along with the fixed intrinsic parameters provided by the dataset. This yields updated extrinsic parameters for each frame, and we discard frames where the camera position diverges by more than 10 mm from the raw average position. We then compute the average value of the remaining frames as the final extrinsic parameters for the subject. Using these recalibrated extrinsic parameters together with the original intrinsic parameters, we perform multi-view face reconstruction. Based on the reconstruction results, we render the dataset, denoted as MVS-XGazeF-NV. Random factors for background augmentation are kept the same, and sample images from these reconstruction methods are shown in Fig. 7.

2) COMPARISON OF RECONSTRUCTION METHODS

We compare multi-view face reconstruction with state-of-the-art single-view methods, 3DDFA [76] and DECA [75]. As a simple baseline, *Landmarks Fitting*, we fit the BFM model [69] to the detected 68 2D facial landmarks to obtain a facial shape, with texture sampled from the RGB values projected from the original image.

We separate the 80 ETH-XGaze training subjects into four folds to perform leave-one-fold-out evaluation with a ResNet-18 model. We then compare the performances of models trained on synthetic data generated by different face reconstruction methods. The average errors are shown in Tab. 4. From the top three rows, we observe that the three

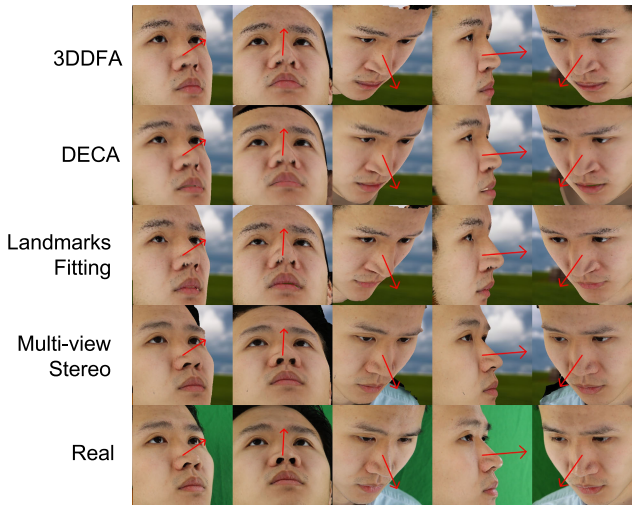


FIGURE 7. Examples of the synthesized XGaze-NV datasets using the different reconstruction methods. The last row shows the samples of the real dataset.

TABLE 4. Comparison of synthetic datasets using different reconstruction methods. The gaze estimation error is the average of the four-fold split. The model used is ResNet18 for all rows. * indicates the green background version without the background augmentation.

Datasets	Within	XGaze-Test
XGazeF-NV-3DDFA	10.03	11.32
XGazeF-NV-DECA	10.02	11.47
XGazeF-NV-fitting	10.28	11.67
XGazeF-NV-3DDFA*	20.33	22.85
MVS-XGazeF-NV *	7.90	7.43
MVS-XGazeF-NV	6.71	6.19
ETH-XGaze	6.62	6.20

single-view methods produce similar performance, which we attribute to their similar appearance and texture quality (resolution). On the other hand, as expected, the high-quality MVS-XGazeF-NV achieves the lowest error, approaching the performance of models trained on real ETH-XGaze (last two rows), which represents an upper bound for synthetic training. Qualitatively, we observe image quality differences between the single-view methods and *Multi-view Stereo* in Fig. 7, such as artifacts in the eyebrow region and reduced resolution. Empirically, such lower image quality is likely to cause the observed performance gap, and future work should provide a more detailed quantitative analysis of how specific artifacts or resolution affect gaze error.

However, although single-view methods perform worse than the multi-view approach, they are easier to deploy in real-world applications without requiring complicated multi-view synchronization. More importantly, there is potential to reduce the gap between single-view and multi-view performance. As shown in the fifth and second rows of Tab. 4, the proposed background augmentation reduces the error of *XGazeF-NV-3DDFA*, demonstrating that appearance diversity is beneficial for generating training data. In addition to background augmentation, further improvements in reconstruction quality, particularly in texture, have the potential to bring single-view reconstruction methods closer to the performance of upper-bound multi-view methods.

VI. CONCLUSION

This work presents an effective data synthesis pipeline and an unsupervised domain adaptation approach for full-face appearance-based gaze estimation. Our approach leverages 3D face reconstruction to synthesize training datasets with larger head poses while preserving accurate gaze labels via projective matching. The proposed DisAE model learns gaze-related features from synthetic data and can therefore generalize effectively to other domains. Moreover, DisAE can be further adapted to target domains through a self-training process with the proposed background-switching consistency loss. Extensive experiments demonstrate that the generated synthetic data benefits model training, and our approach achieves superior performance compared to existing state-of-the-art methods in cross-dataset and unsupervised domain adaptation evaluations. Furthermore, the results verify that synthetic data can achieve performance comparable to real data, highlighting the potential of synthetic training for future research.

LIMITATION

Despite the effectiveness of our approach in extending the gaze range through novel-view rendering using 3D reconstructed faces, it has several limitations. First, the proposed method does not fully explore image appearance augmentation, such as variations in face size, textures, or eye color. This limitation may affect the generalization ability of baseline methods and also reduce the disentangling capability of the proposed model. Future research should therefore focus on synthesizing data that covers a broader spectrum of appearance variations. Addressing these factors will be crucial for further improving the robustness and performance of gaze estimation models.

DATA AVAILABILITY

The reconstruction and synthesis code, along with the updated camera calibrations of ETH-XGaze, are available at https://github.com/ut-vision/Gaze-NV-Rendering

All datasets used in this work were collected with the approval of an IRB or with the consent of the participants [8], [9], [14], [16]. Although the proposed method generates synthetic faces, ethical concerns are minimal because it does not create new identities or extend the diversity of human faces.

DECLARATION OF AI-GENERATED TEXT

During manuscript preparation, the authors used ChatGPT[1] to refine the text and correct grammatical mistakes.

[1] https://chatgpt.com/

All generated content was subsequently reviewed and edited by the authors, who take full responsibility for the final version of the publication.

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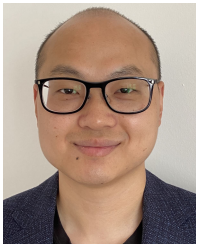
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